

k문량파로 k문량가능 x량로 u라말 멜로 때로검는대로 랑트능는 f 다말말기 두  
x량로

=l xx~|vy=l x|l ukl k U~l y→Sw|p ph ym[ l j |vy/E~nt l u|l k Euhs+ph hsU~l yph z  
=L →uqvu Opt . Gohl yph Pph . L→vuqu Eu \* FHEMPhi + Vh|opp Wu \*Qpyzvnh Kyh→Wz|l t z Phi +  
O±huno→u Thyr \* +  
=hy\_p=4734Q<8<7·4 bj z0HFd  
=4247 34 33

\*PPQ+ VEK\*VI |yph·hs/E~nt l u|l k Kl ul yh|pu+ . .  
OVEK PPQ \* . +  
" \*ohs-j ph|pu+" 0 \*[ WW+ 0  
" |vw/r " 0  
0 " 0  
Euhs+ph hsU~l y+ 0 [ EU\*[ l j |vy/E~nt l u|l k

\*wn·l j |vy. [ FEW . H~jrHF + [ EU  
\* + 0 =  
wn·l j |vy= 550'  
[ FEW = 72'  
H~jrHF= 322' \* . +  
\* = J~ss Xhi d Wy hu. +  
0

l xx~|vy Ol xx~|vy 0  
=  
30l GUS\* xhj | Ghypuhph→U~l y→Sw|p ph|pu+ \*LRW +  
0 0 32kt z  
0  
40 =  
\* x|yhwwsh|l + 0  
0

30[ EU =XTG/L XTG/HW . - 0  
40 = =

wn·l j|vy 3.222  
[ FEW 32.222  
H~j rHF 30 2594

3042504

50 =wn·l j|vy\*Tvz|nyl WUP + [ FEW \*Tvz|nyl WUP + H~j rHF\* HF+  
0  
60 =l GUS 30 : 2308t z 2026' 0  
70 = LRW 0

---

EB

EE yf m

\*PPQ+ . . . VEK\*VI |y p·hs/E~nt l u|l k  
Kl ul yh|pu+ OVEK =  
PPQ \* . . + 0  
PPQ \*vu|l x|+ 0  
PPQ . " \*ohs~j ph|pu+"  
0  
VEK \*l j|vyW p·hy p→W hyj o. [ WW+ 0 =  
30 \*u~t l y p·hs·l j|vy+ 0\* =98: FI VX +  
40 1 \* + 0  
50 PPQ 0

EB

\* +  
Qp~z. T p l j vul . Ukyhu|. ] l h·p|l . Goyt hHF 0  
0 =  
WUP \* . . + 0  
\* + 0  
0

wn·l j|vy\*Tvz|nyl WUP + [ FEW \*Tvz|nyl WUP + Euhs~p HF. H~j rHF/[ WW~j rHF + 0  
=  
0  
WUP . . 0  
[ EU\*[ l j|vy/E~nt l u|l k Euhs~p hsU~l y~+ 0  
0

EB

\*RI hyl z| RI poi vyW hyj o+ =  
ORR\*O/RI hyl z| RI poi vy+  
r 0  
OS\*u~k+ . u . k +  
0



g  
40 0 wz0zfl t i l kkp n ?/B %3 ? 20 0  
50l xx~|vy I GUS =  
\*LRW] + 0  
.%3+ \*20+ LRW] 0  
\* =456 + 0  
60 456 0  
\*Rl z|l k Pvvw·z Lhzo Nvp+  
\*Wkl xW] hu ·z W x~l u|p sW] hu+  
70 0  
80 LRW] 0  
C  
LRW] 0\* 324t z+  
LRW] \*kl |l yt p p|p+ . . . \*l rnzl hyj o +  
0 0  
=  
wn·l j |vy.[ FEW= \*shuul y+ \*ovvr+ 0  
H~jrHF= \*svng]hs vw|p p]l y+ 0 H~jrHF  
\*p/wvj l zz+ 0  
FF  
I GUS 0\* + l xx~|vy  
0  
=  
30 \* + 0  
40 0  
50 0  
\* +=  
N = ceil((z² × P\_hat × (1 - P\_hat)) / e²)  
=  
δ= \*<7' 둘 δ A 308+  
Tfoh|= \* 207. 72' +  
l = \* 207. 7' +  
=  
N = ceil((1.96² × 0.5 × (1 - 0.5)) / 0.05²)  
= ceil((3.8416 × 0.25) / 0.0025)  
= ceil(0.9604 / 0.0025)  
= ceil(384.16)  
= 385  
. 5: 7 <7' 날' 0  
\*Ekhw|p l Wht wqun+=  
=  
=  
=  
l xx~|vy =

```
U/I ywy =
Q-error = max(Card_esti / Card_true, Card_true / Card_esti)
```

U/I ywy 3 .3 0

=

30 U/I ywy \* + 0

40 \* A 2G+ =

g

ful ± A \*3 / t + 낮 f j h j - t 낮 f w y l .

g

50 \* + 0

6072 0

=

HI I T. W t W h y j o R I | - - = \* +

W M X = \* +

GB 감 x T

GB

ERR \* =ERR F l u j o t h y r z + =

= | v w / r 0

= [ E U . . 0

[ E U O X T G / L X T G / H W

0

GB 가 hAn

X T G / L . : 44 0

=

=

why | z ~ w w =

w z f p h n l f l t i l k k p u n =

w z f | l x | f l t i l k k p u n =

w z f | h n = \* +

why w f | l x | f l t i l k k p u n

=

|                              |     |                                                             |
|------------------------------|-----|-------------------------------------------------------------|
|                              |     |                                                             |
| 가////////가////////가////////가 |     |                                                             |
| HI I T                       | <8  |                                                             |
| W M X                        | 34: | * V j h s / M · h y p u l J l h   ~ y l X y h u z n a y t + |
| W t W h y j o R I   - -      | 478 |                                                             |
| ` J G G                      | 3<4 | ` h o v v J g r y G y h l p l G v t t v u z                 |
| ] M                          | 98: |                                                             |

=X T G / L 44 U5. U7. U: . U<. U32. U33. U34. U42 0

GB 4가대능능문대문가대문

=

| v w / r \* y h u n l z l h y j o + =

```
ps_embedding <-> ${image_embedding} < ${D}
=
```

?/B=  
H=

C

Xvw/r = r 0 r . 0 0 205 422 . 207 72.222 0  
= H 0  
0\* ="207 ". " 72rt " +

GBH가hA z

XTG/HW XTG/L =

pl t pl t i l kkpun  
U9. U34. U3<. U42. U64. U94. U<:  
XTG/HW << . [ EU 0

HB

HBE

=M|l s\_l vu Kvsk 8752 \*34: ·GTY+ 30 XF VEQ  
=wn·l j |vy. [ FEW . H~j rHF \*±p o H~j rHF/[ VW+  
=HI I T. WIX. V W hyj oRI |-- . `JGG. ] NM  
=LRW \*QA38. l rñj vuz|y~j |puA422. l rñzl hyj oA622+  
=XTG/L W322 \* : 2 +XTG/HWW32 \* 33 +

HBF khxu T

\*Jp~yl 6+

wn·l j |vy. [ FEW . H~j rHF I xx~|vy kyht h|p p wv·l t l u|=

|                              |                                  |          |
|------------------------------|----------------------------------|----------|
|                              |                                  |          |
| 가////////가////////가////////가 |                                  |          |
| wn·l j  vy                   | 2B.222 *5 vykl yz vnt hnup~kl +  | 3222722  |
| [ FEW                        | 2B2.222 *6 vykl yz vnt hnup~kl + | 72225222 |
| H~j rHF                      | 5904                             | 42B2     |

=wn·l j |vyU5 ±p o HI I T

Exqutor 적용 전:  
Full Table Scan (8000만 행)  
→ Hash Join 3번  
→ 총 실행 시간: 52,535초 (약 14.6시간!)

Exqutor 적용 후:  
HNSW Index Scan (234행 반환)  
→ Nested Loop Join (효율적)  
→ 총 실행 시간: 15.571ms (0.016초)

속도 향상: 52,535 / 0.015571 ≈ 3,368,000배

C

30 = " 456 " 0  
40RI z|l k PvvwNvp = 0  
50 = 0  
60Mkl × Vj hu =W x~l u|p sVj hu Mkl × Vj hu 0  
I GUS \*Xhi s M+

|                              |       |  |  |
|------------------------------|-------|--|--|
|                              | I GUS |  |  |
| 가////////가////////가////////가 |       |  |  |

|                   |          |      |        |
|-------------------|----------|------|--------|
| HI I T            | 30 : t z | 6506 | 20265' |
| WMX               | 30 <t z  | 6307 | 20267' |
| Vp W hyj oRl   -- | 30<8t z  | 6508 | 20267' |

0

H66 T

\*Jpn~yl 7≠

$$\begin{aligned} &=wn\cdot Ij|vy \qquad 304 \quad 2504 \\ &= \qquad \qquad \qquad 306 \\ &= \end{aligned}$$

=

$$\begin{aligned} HI I T. Vp W hyj oRl | -- = \qquad \qquad \qquad \text{둘} \qquad \qquad \qquad * \qquad \qquad \qquad + \\ WMX= \qquad \qquad \qquad \text{둘} \qquad \qquad \qquad * \qquad \qquad \qquad + \\ = \end{aligned}$$

$$\begin{aligned} 1 \qquad \qquad \qquad 4: 295t z \quad . I GUS \quad 324t z \qquad \qquad \qquad . \qquad \qquad \qquad *48269 \quad + \\ 0 \end{aligned}$$

H6H

- \*Jpn~yl 9≠

$$\begin{aligned} `JGG \qquad \qquad \qquad *wzf|hn A )uh|~yl ) \qquad \qquad \qquad + \qquad \qquad \qquad = \\ 52307 \\ *Jpn~yl : . <≠ \end{aligned}$$

=

$$\begin{aligned} why|z~ww \qquad =HI I T \qquad \qquad \qquad * <8 \quad + \\ why| \qquad =] 10M \qquad \qquad \qquad *98: \quad + \\ =3029 \quad 269<06 \end{aligned}$$

XTG/HW \*Jpn~yl 32≠

$$\begin{aligned} XTG/L \qquad \qquad \qquad = \\ 32<08 \end{aligned}$$

H6D

W 3Rl | \*Jpn~yl 34≠

W 3Rl | 0

|                              |          |              |
|------------------------------|----------|--------------|
|                              | I xx~ vy | W 3Rl        |
| 가////////가////////가////////가 |          |              |
|                              |          | I xx~ vy 380 |
|                              | Rv       | `I z * +     |
|                              | * +      | * +          |
|                              | * +      | * +          |
|                              | * +      |              |

H6P 4 뎡뎡뎡EH5

$$Wyhs Jhj|vy \quad 3.32.322 \qquad \qquad \qquad =$$

$$wn\cdot Ij|vy \quad =$$

$$W \quad 3 \quad W \quad 322 \quad 322 \quad \text{둘} \quad 322$$

$$I \, xx\sim|vy \quad =$$

LRW  
W 322 303

OB

OBE

\* 98: ] 0M =  
0  
\* ~yzl vnrkt l uzpruhst \*

OF

=  
ERR \*LRW + 0  
0

OB

30 = 0  
40 =ERR 0  
50 = 0

PB 4y 능구는 결 따른

PBE

HF =  
Qp ~z. Ukyhu|=  
EGSVR. W VJ. LUERR. kpr ERR=ERR  
= 0 0

PBF

i g  
=  
Euhs+g HF=  
Wuns Wvyl =  
= 0 . 0

PBG

=  
=  
l xx~|vy 0

QB

=  
30I GUS= . 6 vykl yz vmt hnup~kl \*32.222 + 0  
32kt z 0  
40 = 304250 0  
0  
50[ EU =XTG/L 1XTG/HW - 0 [ EU  
0  
60 =wn·l j |vy\*Tvz|nyl WUP+ [ FEW \*Tvz|nyl WUP + H~j rHF\* SPET+



|       |           |      |      |        |     |   |          |      |    |
|-------|-----------|------|------|--------|-----|---|----------|------|----|
|       |           | 0    |      | 0      |     |   |          |      |    |
| 70    | =         | [ EU |      | OPPQ - | -   |   |          | 0    |    |
| <hr/> |           |      |      |        |     |   |          |      |    |
| 30    |           | =    |      |        | 0   |   |          |      |    |
|       | .         |      |      | C      |     |   |          | C    |    |
| 40    |           | =]   | MM8: | +      |     |   |          |      | 0  |
|       |           |      |      | C      |     |   |          |      |    |
| 50    |           | =    | "    |        | "   | . |          |      |    |
|       | Ol xx~ vy |      |      | C      |     |   |          |      |    |
| 60    |           | =    |      |        |     | . |          |      | 0  |
| I GUS |           | .    |      | C      |     |   |          |      |    |
| 70    |           | =    |      |        | * = |   | ‡ LRW    |      |    |
|       |           |      |      | C      |     |   |          |      |    |
| 80    |           | =    |      |        | * = |   | ? 20 ERH | ? 20 | 5‡ |
|       |           |      | C    |        |     | 0 |          |      |    |